

Circuit Analysis (Lab # 6)

Experiment 6: VERIFICATION OF KIRCHHOFFS LAWS

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Grade and Signature:

CLO2	Use with identification the circuit components, breadboards, multimeters, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
CLO3				
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objective

To verify Kirchhoff's current law and Kirchhoff's voltage law for the given Circuit.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	330ohm, 220ohm 1kohm	3

Description

KCL: The algebraic sum of the currents meeting at a node is equal to zero.

KVL: In any closed path / mesh, the algebraic sum of all the voltages is zero.

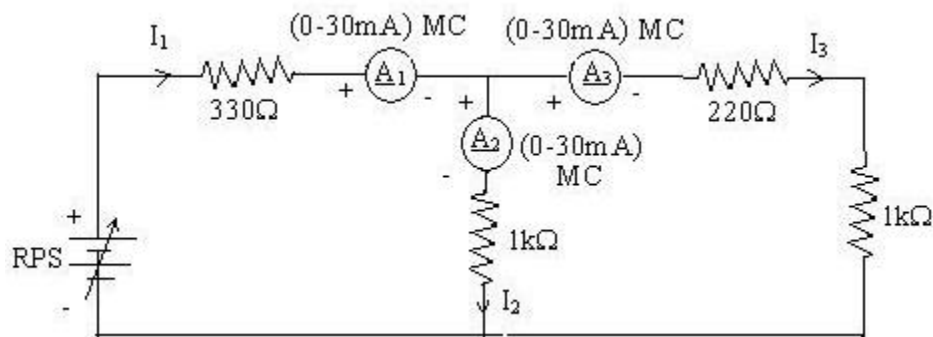
Precautions:

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.

Procedure for KCL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note down the corresponding readings using multimeter
4. Repeat the same for different voltages

Circuit - KCL



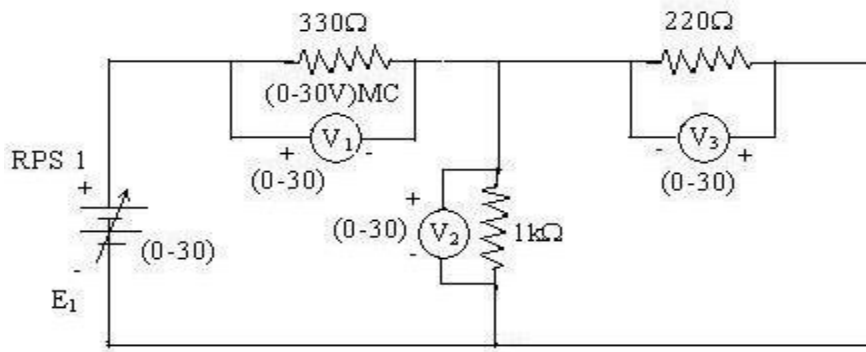
KCL - Practical Values:

Sl. No.	Voltage E Volts	Current			$I_1 = I_2 + I_3$ mA
		I_1 mA	I_2 mA	I_3 mA	
1	5	5.68	3.12	2.56	5.68
2	15	17.05	9.37	7.68	17.05
3	25	28.42	15.62	12.80	28.42

Procedure for KVL:

1. Give the connections as per the circuit diagram.
2. Set a value in RPS.
3. Note all the voltage reading
4. Repeat the same for different voltages

Circuit - KVL



KVL - Practical Values

Sl. No.	RPS	Voltage			$E_1 = V_1 + V_2$
	E_1	V_1	V_2	V_3	
	V	V	V	V	
1	5	3.23	1.77	1.77	5.00
2	10	6.47	3.53	3.53	10.00
3	15	9.70	5.30	5.30	15.00

Result:

Thus, Kirchhoff's voltage law and Kirchhoff's current law are verified practically by performing this experiment.

Post Lab Questions

1. Define Kirchhoff's current law.

Kirchhoff's current law states that the total current entering a junction is equal to the total current leaving it. The net current at any electrical node is exactly zero.

2. Define Kirchhoff's voltage law.

Kirchhoff's voltage law states that the total sum of all voltages around any closed loop in a circuit must equal zero. The total voltage supplied is always equal to the total voltage dropped across the components.

3. What are the applications of regulated power supply?

Regulated power supplies are used to provide a steady DC voltage for sensitive electronic devices. Common applications include computer power supplies and mobile phone chargers. They are also used in laboratory test equipment to make sure readings are accurate.

4. What are the limitations of KCL?

KCL assumes that no electrical charge builds up at a junction over time. This means it only works well for standard circuit models. It is not valid for very high frequency circuits where radiation and stray capacitance become major factors.
